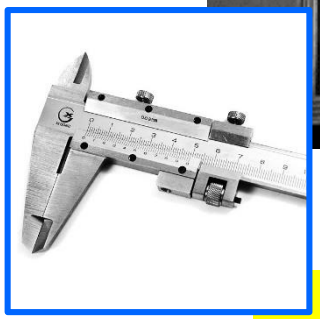
Stony Brook Institute at Anhui University

安徽大学纽约石溪学院

Chapter 2 Random variable and their probability distributions

Section 2.9 Probability distribution of functions of two random variables



$$g = \frac{4l\pi^2}{T^2}$$

Question:

- T needs to be measured several times and get a average to make the measurement of T more accuracy.
- The length l of this clock also needs to be measured several times to get a average value.
- T is a Normal distribution. L is a Normal distribution.

➢ So, what is the distribution of g?

➢ **How to find the probability distribution of random variable Z = f (X,Y), according to the probability distribution of X and Y.**

Section 2.9 Probability distribution of functions of two random variables

- Probability distribution of functions of two discrete random variables

- Probability distribution of functions of two continuous random variables

Probability distribution of functions of two discrete random variables

Example 2.9.1: The distribution law of r.v. (X,Y) is

$X \backslash Y$	-2	-1	0
-1	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{3}{12}$
$\frac{1}{2}$	$\frac{2}{12}$	$\frac{1}{12}$	0
3	$\frac{2}{12}$	0	$\frac{2}{12}$

Find the distribution laws of (1)X+Y, (2) |X-Y|

Solution: (1)

P	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{3}{12}$	$\frac{2}{12}$	$\frac{1}{12}$	$\frac{2}{12}$	$\frac{2}{12}$
(X,Y)	$(-1,-2)$	$(-1,-1)$	$(-1,0)$	$\left(\frac{1}{2},-2\right)$	$\left(\frac{1}{2},-1\right)$	$(3,-2)$	$(3,0)$

Probability distribution of functions of two discrete random variables

P	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{3}{12}$	$\frac{2}{12}$	$\frac{1}{12}$	$\frac{2}{12}$	$\frac{2}{12}$
(X,Y)	$(-1,-2)$	$(-1,-1)$	$(-1,0)$	$\left(\frac{1}{2},-2\right)$	$\left(\frac{1}{2},-1\right)$	$(3,-2)$	$(3,0)$
$X+Y$	-3	-2	-1	$-\frac{3}{2}$	$-\frac{1}{2}$	1	3
$\lvert X-Y\rvert$	1	0	1	$\frac{5}{2}$	$\frac{3}{2}$	5	3

Probability distribution of functions of two discrete random variables

So the distribution law of X+Y and |X-Y| are:

$X+Y$	-3	-2	-1	$-\frac{3}{2}$	$-\frac{1}{2}$	1	3
P	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{3}{12}$	$\frac{2}{12}$	$\frac{1}{12}$	$\frac{2}{12}$	$\frac{2}{12}$

$\lvert X-Y\rvert$	0	1	$\frac{5}{2}$	$\frac{3}{2}$	5	3
P	$\frac{1}{12}$	$\frac{4}{12}$	$\frac{2}{12}$	$\frac{1}{12}$	$\frac{2}{12}$	$\frac{2}{12}$

Probability distribution of functions of two discrete random variables

Conclusion:

If the distribution law of r.v. (X,Y) is

$$P\{X = x_i, Y = y_j\} = p_{ij}, i, j = 1,2,\cdots$$

Then the distribution law of r.v. $Z = f(X,Y)$ is given by

$$P\{Z = z_k\} = P\{f(X,Y) = z_k\}$$

$$= \sum_{z_k = f(x_i, y_j)} p_{ij} \; k = 1,2,\cdots.$$

Probability distribution of functions of two discrete random variables

Suppose that the distribution law of two independent random variables X and Y are:

X	1	3
P_X	0.3	0.7

Y	2	4
P_Y	0.6	0.4

Find the distribution laws Z=X+Y and Z=X·Y.

Solution: due to X and Y are independent from each other, so :

$$P\{X = x_i, Y = y_j\} = P\{X = x_i\}P\{Y = y_j\},$$

X \ Y	2	4
1	0.18	0.12
3	0.42	0.28

Probability distribution of functions of two discrete random variables

$X \backslash Y$	2	4
1	0.18	0.12
3	0.42	0.28

P	(X,Y)	$Z=X+Y$	$Z=X\bullet Y$
0.18	(1,2)	3	2
0.12	(1,4)	5	4
0.42	(3,2)	5	6
0.28	(3,4)	7	12

$Z=X+Y$	3	5	7
P	0.18	0.54	0.28

$Z=X\bullet Y$	2	4	6	12
P	0.18	0.12	0.42	0.28

Probability distribution of functions of two discrete random variables

Example 2.9.2: The distribution law of two independent random variables are the same and given by

X	0	1
P	0.5	0.5

Find the distribution law of Z=MAX(X,Y).

Solution: X and Y are independent, so :

$$P\{X = x_i, Y = y_j\} = P\{X = x_i\}P\{Y = y_j\},$$

$X \backslash Y$	0	1
0	$1/2^2$	$1/2^2$
1	$1/2^2$	$1/2^2$

Probability distribution of functions of two discrete random variables

$$P\{\max(X,Y)=i\}$$

$$=P\{X=i,Y<i\}+P\{X\le i,Y=i\}$$

$X \backslash Y$	0	1
0	$1/2^2$	$1/2^2$
1	$1/2^2$	$1/2^2$

$$P\{\max(X,Y)=0\}=P\{0,0\}=\frac{1}{2^2},$$

$$P\{\max(X,Y)=1\}=P\{1,0\}+P\{0,1\}+P\{1,1\}=\frac{3}{2^2}.$$

So, the distribution law of Z=MAX(X,Y) as:

Z	0	1
P	$\frac{1}{4}$	$\frac{3}{4}$

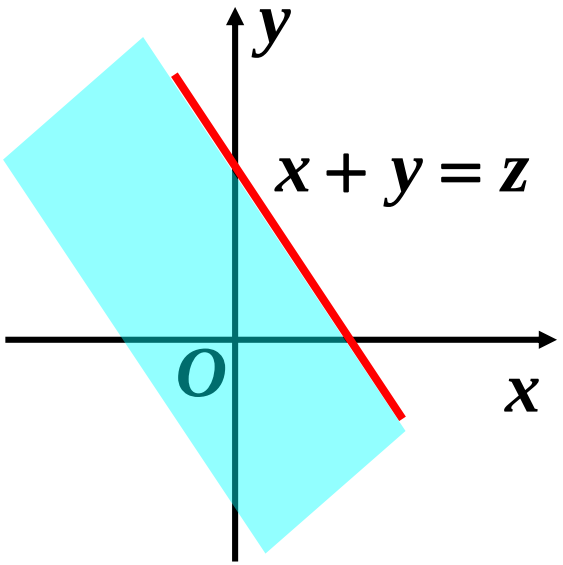
Section 2.9 Probability distribution of functions of two random variables

- Probability distribution of functions of two discrete random variables
- Probability distribution of functions of two continuous random variables

Probability distribution of functions of two continuous random variables

1. The distribution of Z=X+Y

If the joint PDF of r.v. X and Y is $f(x,y)$, then the PDF of $Z=X+Y$ is:



$$F_Z(z) = P\{Z \le z\} = \iint\limits_{x+y\le z} f(x,y)\,\mathrm{d}x\,\mathrm{d}y$$

$$= \int_{-\infty}^{\infty}\left[\int_{-\infty}^{z-y} f(x,y)\,\mathrm{d}x\right]\mathrm{d}y$$

$$\xlongequal{x=u-y} \int_{-\infty}^{\infty}\left[\int_{-\infty}^{z} f(u-y,y)\,\mathrm{d}u\right]\mathrm{d}y$$

$$= \int_{-\infty}^{z}\left[\int_{-\infty}^{\infty} f(u-y,y)\,\mathrm{d}y\right]\mathrm{d}u.$$

Probability distribution of functions of two continuous random variables

$$F_Z(z) = P\{Z \le z\} = \int_{-\infty}^{z} [\int_{-\infty}^{\infty} \mathrm{f}(u - y, y)\, \mathrm{d}\, y]\, \mathrm{d}\, u.$$

So the PDF of Z is: $$f_Z(z) = \int_{-\infty}^{\infty} \mathrm{f}(z - y, y)\, \mathbf{d}\, y\,.$$

Similarly,

$$f_Z(z) = \int_{-\infty}^{\infty} \mathrm{f}(x, z - x)\, \mathbf{d}\, x\,.$$

When X and Y are independent,

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(z - y) \mathrm{f}_Y(y)\, \mathbf{d}\, y\,,$$

$$f_Z(z) = \int_{-\infty}^{\infty} \mathrm{f}_X(x) f_Y(z - x)\, \mathbf{d}\, x\,.$$

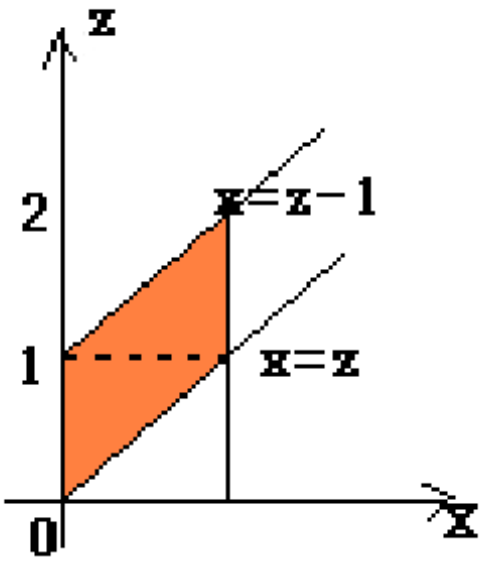
Probability distribution of functions of two continuous random variables

Example 2.9.3: The PDF of Independent r.v. X and Y are both

$$f(x) = \begin{cases} 1\,, 0 \le x \le 1 \\ 0, \qquad others \end{cases}$$

Find the PDF of **Z=X+Y**.

Solution: $$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x)dx$$



The limit of integration is

$$\begin{cases} 0 \le x \le 1 \\ 0 \le z - x \le 1 \end{cases} \Rightarrow \begin{cases} 0 \le x \le 1 \\ x \le z \le x+1 \end{cases}$$

$$f_Z(z) = \begin{cases} \int_0^z dx = z, & 0 \le z < 1 \\ \int_{z-1}^1 dx = 2 - z, & 1 \le z < 2 \\ 0, & others \end{cases}$$

Probability distribution of functions of two continuous random variables

Example 2.9.4: Independent r.v. X and Y both have Normal distribution, and $X \sim N(\mu_1, \sigma_1^2)$ $\quad Y \sim N(\mu_2, \sigma_2^2)$

Find the PDF of Z=X+Y.

Solution:

$$f_Z(z) = \int_{-\infty}^{+\infty} f_X(x)\, f_Y(\underline{z-x})\, dx$$

$$= \int_{-\infty}^{+\infty} \left(\frac{1}{\sqrt{2\pi}\sigma_1} e^{-\frac{(x-\mu_1)^2}{2\sigma_1^2}}\right) \cdot \left(\frac{1}{\sqrt{2\pi}\sigma_2} e^{-[(\underline{z-x})-\mu_2]^2 / 2\sigma_2^2}\right) dx$$

$$= \frac{1}{2\pi\sigma_1\sigma_2} \int_{-\infty}^{+\infty} e^{-\frac{(x-\mu_1)^2}{2\sigma_1^2} - \frac{(z-x-\mu_2)^2}{2\sigma_2^2}}\, dx$$

Probability distribution of functions of two continuous random variables

$$= \frac{1}{2\pi\sigma_1\sigma_2}\int_{-\infty}^{+\infty} e^{-\frac{(x-\mu_1)^2}{2\sigma_1^2}-\frac{(z-x-\mu_2)^2}{2\sigma_2^2}}\,dx$$

$$= \frac{1}{2\pi\sigma_1\sigma_2}\cdot e^{-\frac{(z-\mu_1-\mu_2)^2}{2(\sigma_1^2+\sigma_2^2)}}\int_{-\infty}^{+\infty} e^{-\frac{\sigma_1^2+\sigma_2^2}{2\sigma_1^2\sigma_2^2}\left(x-\frac{\sigma_2^2\mu_1-\sigma_1^2\mu_2+\sigma_1^2 z}{\sigma_1^2+\sigma_2^2}\right)}\,dx$$

$$= \frac{1}{\sqrt{2\pi}\sqrt{\sigma_1^2+\sigma_2^2}}\,e^{-\frac{(z-\mu_1-\mu_2)^2}{2(\sigma_1^2+\sigma_2^2)}}$$

Let $t=\frac{\sqrt{\sigma_1^2+\sigma_2^2}}{\sigma_1\sigma_2}\left(x-\frac{\sigma_2^2\mu_1-\sigma_1^2\mu_2+\sigma_1^2 z}{\sigma_1^2+\sigma_2^2}\right)$, then $dt=\frac{\sqrt{\sigma_1^2+\sigma_2^2}}{\sigma_1\sigma_2}\,dx$

$$Z=X+Y\sim N(\mu_1+\mu_2,\sigma_1^2+\sigma_2^2)$$

Probability distribution of functions of two continuous random variables

Note:

- Generally, if r.v. X and Y are independent and $X \sim N(\mu_1, \sigma_1^2)$, $Y \sim N(\mu_2, \sigma_2^2)$, then Z=X+Y is still normal distribution, and $Z \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$.

- It is extended to the sum of N mutually independent normal random variables. Suppose $X_i \sim N(\mu_i, \sigma_i^2) \quad i = 1, 2, 3, \cdots n$,and their independent of each other. Then the sum of X_i still be a normal distribution.

$$Z = X_1 + X_2 + \cdots X_n \sim N(\mu_1 + \mu_2 + \cdots \mu_n, \sigma_1^2 + \sigma_2^2 + \cdots \sigma_n^2)$$

- Furthermore, the linear combination of a finite number of independent normal random variables still be a normal distribution.

Probability distribution of functions of two continuous random variables

Example 2.9.5: Independent r.v. X and Y both have Poission distribution, $X \sim P(\lambda_1), Y \sim P(\lambda_2)$.
Find the PDF of Z=X+Y.

Solution:

Because X and Y are take values in 0,1,2,…,
So, Z take values in nonnegative integers k, thus:

$$
\begin{aligned}
P(Z=k) &= P(X+Y=k) \\
&= P(X=0, Y=k) + P(X=1, Y=k-1) \\
&\quad + \cdots + P(X=k, Y=0) \\
&= P(X=0)P(Y=k) + P(X=1)P(Y=k-1) \\
&\quad + \cdots + P(X=k)P(Y=0)
\end{aligned}
$$

Probability distribution of functions of two continuous random variables

$$= e^{-\lambda_1}\frac{\lambda_2^k}{k!}e^{-\lambda_2} + \frac{\lambda_1}{1!}e^{-\lambda_1}\cdot\frac{\lambda_2^{k-1}}{(k-1)!}e^{-\lambda_2} + \cdots + \frac{\lambda_1^k}{k!}e^{-\lambda_1}e^{-\lambda_2}$$

$$= \frac{1}{k!}e^{-(\lambda_1+\lambda_2)}\cdot[\lambda_2^k + \frac{k}{1!}\lambda_1\lambda_2^{k-1} + \cdots + \lambda_1^k]$$

$$= \frac{1}{k!}e^{-(\lambda_1+\lambda_2)}\cdot(\lambda_1+\lambda_2)^k \quad = \frac{(\lambda_1+\lambda_2)^k}{k!}e^{-(\lambda_1+\lambda_2)}$$

Note: If X and Y obey the Poisson distribution with parameters $\lambda_1$and λ_2, X and Y are independent of each other at the same time. Then their sum still obeys Poisson distribution with the parameters (λ_1+λ_2), namely:

$$Z = X+Y \sim P(\lambda_1+\lambda_2)$$

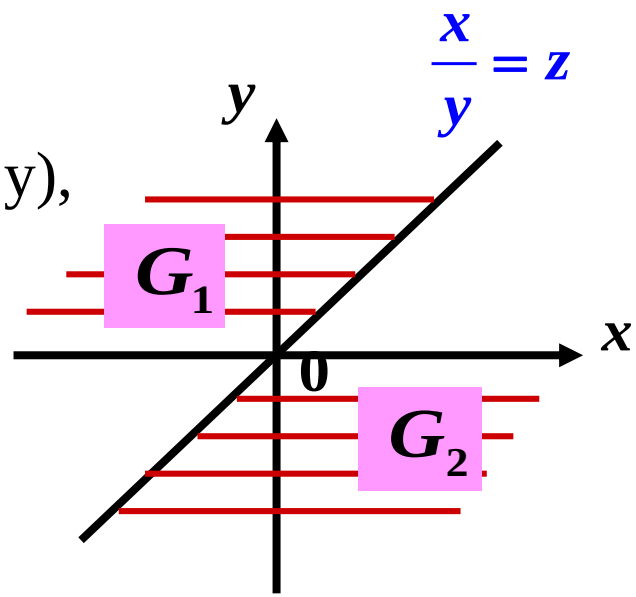
Probability distribution of functions of two continuous random variables

- Examples show that if continuous random variables or discrete random variables obey normal distribution, Γ distribution or Poisson distribution, their sum still obeys normal distribution, Γ distribution or Poisson distribution. And the parameters are the addition of single parameters. Random variables with this property are also called the additive random variables.
- In the process of solving examples, it can be seen that when calculating the distribution of the function $Z=g(X, Y)$, the key point is try to convert it into the form that (x, y) takes value within a certain range, so we can calculate the distribution of Z by using the give distribution of joint (X,Y).

Probability distribution of functions of two continuous random variables

The distribution of $Z = \frac{X}{Y}$

If the joint PDF of r.v. X and Y is f(x,y), then the pdf of Z=X/Y is:



$$F_Z(z) = P(Z \le z) = \iint_{G_1} + \iint_{G_2}$$

For G_1 :

$$\iint_{G_1} f(x,y)\,dx\,dy = \int_0^{+\infty} dy \int_{-\infty}^{yz} f(x,y)\,dx$$

$$= \int_0^{+\infty} dy \int_{-\infty}^{z} y\,f(yu,y)\,du$$

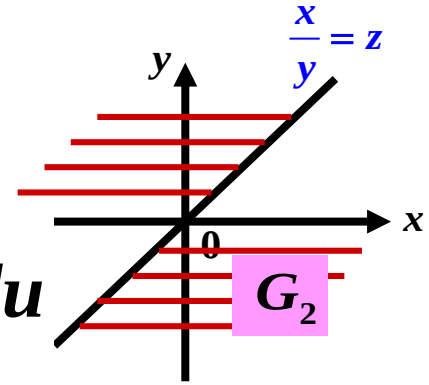
$$= \int_{-\infty}^{z} \int_0^{+\infty} y\,f(yu,y)\;dy\,du$$

fix z, y ,and let

$u = x/y \quad (y > 0)$

Probability distribution of functions of two continuous random variables

Similarly, for G_2: Let $u = \frac{x}{y}$ $(y < 0)$



$$\iint_{G_2} f(x,y)\,dxdy = -\int_{-\infty}^{z}\int_{-\infty}^{0} y\,f(yu,y)\,dydu$$

$$F_z(z) = \iint_{G_1} + \iint_{G_2} = \int_{-\infty}^{z}\int_{0}^{+\infty} y\,f(yu,y)\;dy\,du$$

$$= \int_{-\infty}^{z}\left[\int_{0}^{+\infty} y\,f(yu,y)\,dy - \int_{-\infty}^{0} y\,f(yu,y)\,dy\right]du$$

$$f_Z(z) = \int_{0}^{+\infty} y\,f(yz,y)\,dy - \int_{-\infty}^{0} y\,f(yz,y)\,dy$$

$$= \int_{-\infty}^{+\infty} |y|\,f(yz,y)\,dy$$

When X and Y are independent,

$$f_Z(z) = \int_{-\infty}^{+\infty} |y| \cdot f_X(yz) \cdot f_Y(y)\,dy$$

Probability distribution of functions of two continuous random variables

Nonlinear function $Z = Y/X$ and $Z = XY$

Theorem Let (X, Y) be a continuous two-dimensional r.v., and its joint PDF is $f(x, y)$, then $Z = Y/X$ and $Z = XY$ are continuous random variables, and its PDF are

$$f_{\frac{Y}{X}}(z) = \int_{-\infty}^{+\infty} |x| f(x, \mathrm{xz}) \mathrm{d}x = \int_{-\infty}^{+\infty} |y| f(\mathrm{zy}, y) \mathrm{d}y$$

and

$$f_{XY}(z) = \int_{-\infty}^{+\infty} \frac{1}{|x|} f\left(x, \frac{z}{x}\right) dx = \int_{-\infty}^{+\infty} \frac{1}{|y|} f\left(\frac{z}{y}, y\right) dy$$

Probability distribution of functions of two continuous random variables

Nonlinear function $Z = Y/X$ and $Z = XY$

If X and Y are independent, then $f(x, y) = f_X(x) f_Y(y)$, so

$$f_{\frac{Y}{X}}(z) = \int_{-\infty}^{+\infty} |x| f_X(x) f_Y(\mathrm{xz}) \mathrm{d}x$$

$$= \int_{-\infty}^{+\infty} |y| f_X(\mathrm{zy}) f_Y(y) \mathrm{d}y$$

and

$$f_{\mathrm{XY}}(z) = \int_{-\infty}^{+\infty} \frac{1}{|x|} f_X(x) f_Y\left(\frac{z}{x}\right) \mathrm{d}x$$

$$= \int_{-\infty}^{+\infty} \frac{1}{|y|} f_X\left(\frac{z}{y}\right) f_Y(y) \mathrm{d}y$$

Probability distribution of functions of two continuous random variables

Example 2.9.5: X and Y are the lifetime of two different type of lamps, X and Y are independent. The PDF of X and Y are

$$f(x)=\begin{cases} e^{-x}, & x>0 \\ 0, & others \end{cases} \text{ and } f(y)=\begin{cases} 2e^{-2y}, & y>0 \\ 0, & others \end{cases}.$$

Find the PDF of $Z=\frac{X}{Y}$.

Solution:

When $z \le 0$,

$$f_Z(z)=\int_{-\infty}^{+\infty}|y|\ f_X(yz)\ f_Y(y)\,dy\ =0$$

When $z > 0$,

$$f_Z(z)=\int_{-\infty}^{+\infty}|y|\ f_X(yz)\ f_Y(y)\,dy$$

$$=\int_{-\infty}^{0}(-y)\cdot 0\,dy+\int_{0}^{+\infty}y\cdot e^{-yz}\cdot 2e^{-2y}\,dy$$

Probability distribution of functions of two continuous random variables

When $z > 0$,

$$f_Z(z) = \int_{-\infty}^{+\infty} |y| \; f_X(yz) \; f_Y(y)\, dy$$

$$= \int_{-\infty}^{0} (-y)\cdot 0\, dy + \int_{0}^{+\infty} y \cdot e^{-yz} \cdot 2e^{-2y}\, dy$$

$$= 2\int_{0}^{+\infty} y e^{-(2+z)y} dy = \frac{2}{(2+z)^2}$$

$$\Rightarrow \quad f_Z(z) = \begin{cases} \dfrac{2}{(2+z)^2} & z > 0 \\ 0 & z \le 0 \end{cases}$$

Probability distribution of functions of two continuous random variables

The distribution of max(X,Y) or min(X,Y)

Nonlinear function $M = \max\{X, Y\}$

Suppose X and Y are independent random variables, their cumulative distribution functions are $F_X(x)$ and $F_Y(y)$. So

$$F_{\mathrm{M}}(z) = P\{\mathrm{M} \leq z\}$$
$$= P\{\max\{X, Y\} \leq z\} = P\{X \leq z, Y \leq z\}$$
$$= P\{X \leq z\}P\{Y \leq z\} = F_X(z)F_Y(z)$$

Then

$$F_M(z) = F_X(z)F_Y(z)$$

Probability distribution of functions of two continuous random variables

Nonlinear function $N = \min\{X, Y\}$

Suppose X and Y are independent random variables, their cumulative distribution functions are $F_X(x)$ and $F_Y(y)$. Similarly,

$$F_{\mathrm{N}}(z) = P\{\mathrm{N} \le z\} = P\{\min\{X, Y\} \le z\}$$

$$= 1 - P\{\min\{X, Y\} > z\} = 1 - P\{X > z, Y > z\}$$

$$= 1 - P\{X > z\}P\{Y > z\}$$

$$= 1 - [1 - P\{X \le z\}][1 - P\{Y \le z\}]$$

$$= 1 - [1 - F_X(z)][1 - F_Y(z)]$$

Then

$$F_N(z) = 1 - [1 - F_X(z)][1 - F_Y(z)]$$

Probability distribution of functions of two continuous random variables

General case for extrema

Suppose $X_1, X_2, \cdots, X_n$ are independent random variables, their cumulative distribution functions are $F_{X_i}(x_i)(i = 1, 2, \cdots, n)$. Let $M = \max\{X_1, X_2, \cdots, X_n\}$ and $N = \min\{X_1, X_2, \cdots, X_n\}$, by similar argument as above, we have

$$F_M(z) = F_{X_1}(z)F_{X_2}(z)\cdots F_{X_n}(z)$$

and

$$F_N(z) = 1 - [1 - F_{X_1}(z)][1 - F_{X_2}(z)]\cdots[1 - F_{X_n}(z)]$$

If $X_1, X_2, \cdots, X_n$ have the same cumulative distribution function $F(x)$, then we have $F_M(z) = [F(z)]^n$ and $F_N(z) = 1 - [1 - F(z)]^n$.

Probability distribution of functions of two continuous random variables

Example: Suppose that random variables X_1 and X_2 are independent of each other and have the same geometric distribution, that is:

$$P(X_i = k) = p(1-p)^{k-1}, \quad k = 1, 2, \cdots \quad (i = 1, 2)$$

Find the distribution of $Y = \max(X_1, X_2)$.

Solution:

Method 1#

$$P(Y = n) = P(\max(X_1, X_2) = n)$$

$$= P(X_1 = n, X_2 \le n) + P(X_2 = n, X_1 < n)$$

Probability distribution of functions of two continuous random variables

$$P(\,Y = n\,) = P(\,\max(X_1, X_2) = n\,)$$

$$= P(X_1 = n,\ X_2 \le n\,) + P(\,X_2 = n, X_1 < n\,)$$

Let $1 - p = q$

$$= pq^{n-1}\sum_{k=1}^{n} pq^{k-1} + pq^{n-1}\sum_{k=1}^{n-1} pq^{k-1}$$

$$= p^2 q^{n-1}\frac{1-q^n}{1-q} + p^2 q^{n-1}\frac{1-q^{n-1}}{1-q}$$

$$= pq^{n-1}(2 - q^n - q^{n-1})$$

$$n = 0, 1, 2, \ldots$$

Probability distribution of functions of two continuous random variables

Method 2#

$$P(Y=n)=P(Y\le n)-P(Y\le n-1)$$

$$=P(\max(X_1,X_2)\le n)-P(\max(X_1,X_2)\le n-1)$$

$$=P(X_1\le n,\ X_2\le n)-P(X_1\le n-1,\ X_2\le n-1)$$

$$=[\sum_{k=1}^{n}pq^{k-1}]^2-[\sum_{k=1}^{n-1}pq^{k-1}]^2$$

$$=p^2[\frac{1-q^n}{1-q}]^2-p^2[\frac{1-q^{n-1}}{1-q}]^2$$

$$=(1-q^n)^2-(1-q^{n-1})^2$$

$$=pq^{n-1}(2-q^n-q^{n-1})\qquad n=0,1,2,\ldots$$